Student Name: _____

Student ID# _____

	Check which tutorial for pick-up
T03 Tuesdays 2:30 – 4:20	
T01 Fridays 9:30 – 11:20	

CHEMICAL ENGINEERING 2O04
Fluid Mechanics -Midterm #1

DAY CLASS
Duration 120 minutes (6:00pm – 8:00pm) **D.R. Latulippe**
McMaster University Term Test **Friday February 8th, 2019**

The test includes **7 questions and 1 bonus question**; ensure your copy of the test is complete.

SPECIAL INSTRUCTIONS

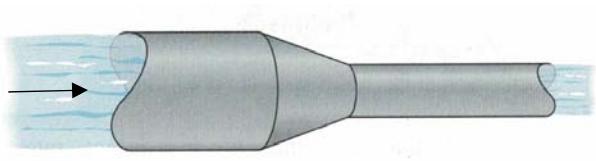
1. The class **textbook (any version of the Pritchard's <u>Fox & McDonald</u>) and any notes** are allowed.
2. McMaster calculator only (and case with unit conversions) is permitted.
3. Only tests written in pen will be considered if re-marking is requested (addition errors excepted).
4. On the last page, I have posted some useful tables from the Appendix of the 9th edition textbook.
5. Marks for each problem are given (**Total of 46 marks)**. Use this as a time-management guide; for example, a question worth only 1 marks should take less than 3 minutes:

$$1 mark \left(\frac{120\ minutes}{49\ marks}\right) = 2.6\ minutes$$

QUESTION 1 - 10 MARKS

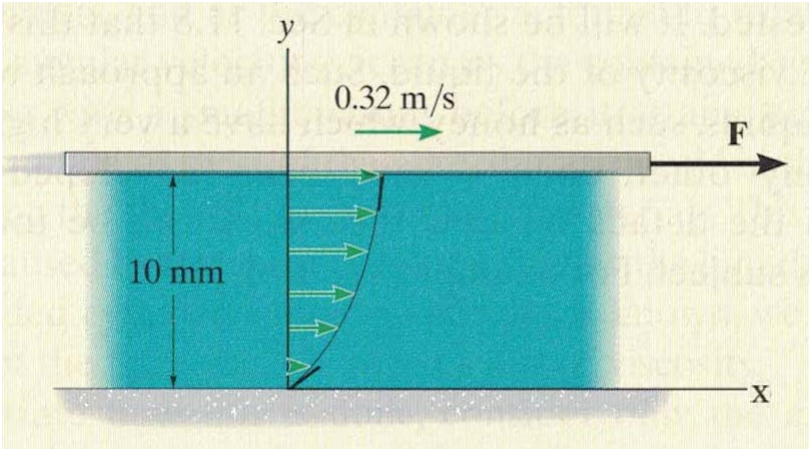
Answer the following short questions in the space provided:

a.	The air pressure within a bicycle tire is determined from a gauge to be 10 psi. If the local atmospheric pressure is 14.6 psi, determine the absolute pressure (in psi) in the tire. **(1 mark)**	
b.	In class we discussed how a mixture of cornstarch in water is a classic non-Newtonian fluid. Give a brief explanation of how this material behaves. **(1 mark)** Sketch the appropriate relationship between shear stress and shear rate for this material. **(2 marks)**	

c.	In tutorial #3, I gave you the following equation for Reynolds number $$Re = \frac{\rho V D}{\mu}$$ Re-write this equation using kinematic viscosity instead of absolute viscosity and density. **(1 mark)**	
d.	As a liquid flows through the constriction into the smaller pipe, its pressure will decrease. Give a brief explanation for why this phenomenon occurs. **(1 mark)**  If the liquid is water at 40 °C, state the lowest absolute pressure it can reach in the constriction before it begins to boil. **(1 mark)**	
e.	Give a brief explanation of the features of a gate valve and what they are primarily used for in a piping system. **(2 marks)**	
f.	A small diameter tube is inserted into a liquid whose contact angle is 110°. Will the level of liquid in the tube be higher or lower than the level of the liquid outside the tube? **(1 mark)**	

QUESTION 2 – 6 MARKS

As shown in the accompanying figure, a thin flat plate rests on top of a film of ambient temperature (i.e. 20 ºC) water. When a small force **F** is applied to this plate, the velocity profile within the water film can be described as $v_x = 40y - 800y^2$
where y is the distance (m) from the bottom stationary plate.

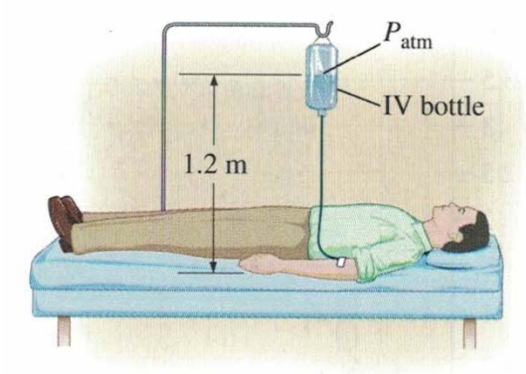


a. Show that velocity equation satisfies the 'no-slip' conditions that must exist for both the top moving plate and bottom stationary plate.
(2 marks)

b. Calculate the magnitude of the shear stress (in Pa) that is acting on the top plate
(4 marks)

QUESTION 3 – 5 MARKS

Intravenous (IV) infusions are usually driven by gravity by hanging a fluid bottle at a sufficient height to counteract the blood pressure of the patient and thus force the fluid into the body.

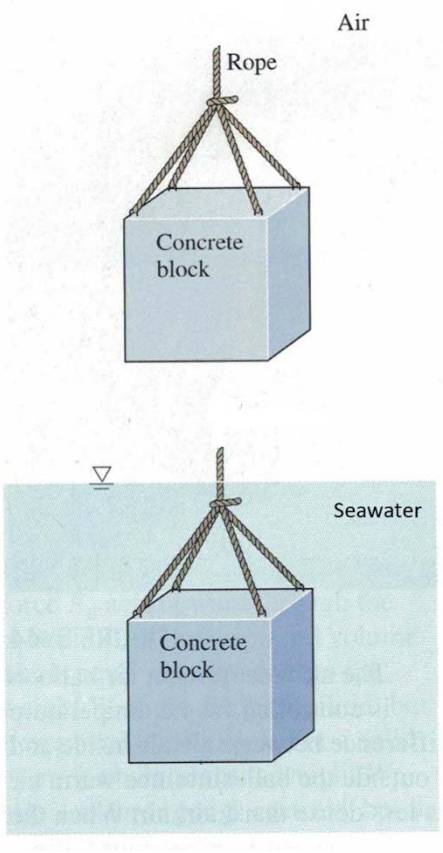


a. A patient is to be given a saline solution (with specific gravity = 1.02) via an IV bottle that is 1.2 m above the injection point in the arm. If it is observed that the pressure of the saline solution at the injection point and the average blood pressure are the same such that no fluid is going into the patient, what must be the blood pressure reading (in kPa) of the patient?
(3 marks)

b. In order to obtain the desired flow rate of saline solution, it was determined that the pressure of the saline solution at the injection point needs to be 8 kPa higher than the answer from part a. To what height (in m) above the injection point in the arm should the IV bottle be placed to achieve this condition?
(2 marks)

QUESTION 4 – 7 MARKS

A crane is used to lower rectangular concrete blocks (0.4 m × 0.4 m × 0.3 m) into the ocean so that they can be used as weights for an underwater construction project.

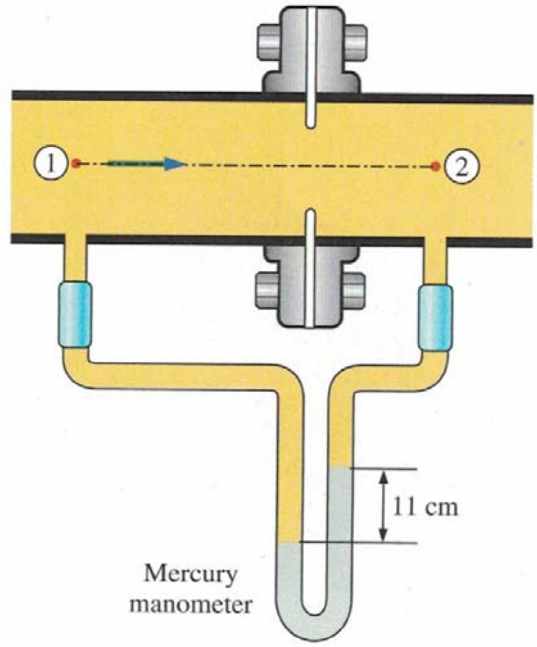


a. Determine the tension in the rope of the crane (in N) when the concrete block is not moving and fully submerged in seawater (at 20 ºC).
(4 marks)

b. Determine the percentage by which the tension in the rope increases when the concrete block is not moving and instead suspended in air (at 10 ºC).
(3 marks)

QUESTION 5 - 10 MARKS

The flow rate of methanol at 20 °C through a horizontal pipe (internal diameter = 4 cm) is to be measured using an orifice meter (internal 'throat' diameter = 3 cm) equipped with a mercury manometer as shown in the accompanying figure.



a. If the differential height of the two legs of the manometer is 11 cm, determine the pressure difference (in Pa) across the orifice meter (i.e. $P_1 - P_2$).
(3 marks)

b. Given that the appropriate equation for the flow rate (Q) through an orifice meter is:

$$Q = A_o C_d \sqrt{\frac{2(P_1 - P_2)}{\rho(1 - \beta^4)}}$$

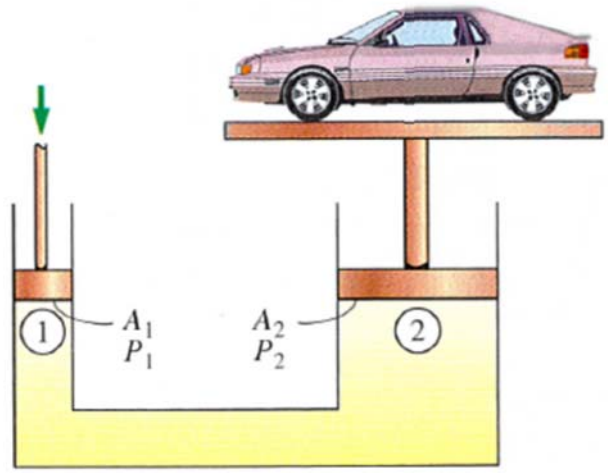
where A_o is the cross-sectional area of the internal 'throat' opening of the orifice meter
C_d is a dimensionless correction factor with an approximate value of 0.61
ρ is the density of the fluid passing through the orifice meter
β is the ratio of internal 'throat' diameter of the orifice meter to internal diameter of the pipe

Determine the flow rate (in m^3/s) of methanol through the horizontal pipe.
(5 marks)

c. Determine the average velocity (in m/s) of methanol in the horizontal pipe where the internal diameter = 4 cm.
(2 marks)

QUESTION 6 - 4 MARKS

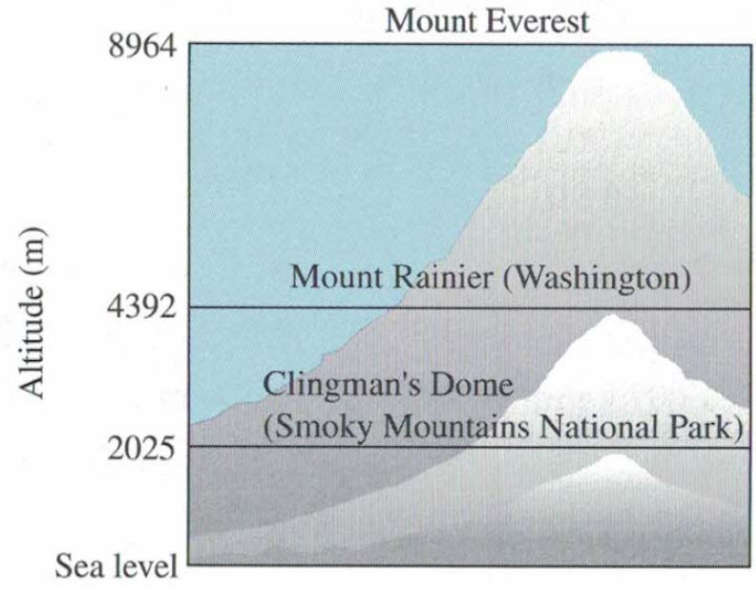
A small car (mass = 1325 kg) is to be lifted using the hydraulic jack (filled with oil with a specific gravity of 0.87) as shown in the accompanying diagram (not drawn to scale); the area of the two pistons are $A_1 = 0.8\ cm^2$ and $A_2 = 400\ cm^2$. When both pistons are located at the same vertical position, determine the magnitude of the force (in N) that is required on the smaller piston to balance the weight of the car.



QUESTION 7 - 4 MARKS

Mount Rainier is the highest mountain in the U.S. state of Washington; it rises 4392 m above sealevel (which is approximately half the height of Mount Everest). If the ambient temperature is 15 ºC at sealevel, determine the absolute pressure at the top of Mount Rainier.

HINT: The universal gas constant in standard SI units is 8314.3 J/kmol·K



BONUS QUESTION – 3 MARKS

The process of adsorption involves separation of a substance from a liquid (or gas) phase accompanied by its accumulation or concentration onto the surface of a solid material. A variety of models have been proposed to describe the mathematical relationship between the equilibrium amount of solute in the liquid phase (C_E) and the equilibrium amount of solute on the solid phase (q_E). The two most common models are:

- The Langmuir model which has the form $q_E = \frac{K \cdot q_{E,max} \cdot C_E}{1 + K \cdot C_E}$ (with two fitting parameters K and $q_{E,max}$) which can be linearized into the form $\frac{1}{q_E} = \frac{1}{K \cdot q_{E,max}} \cdot \frac{1}{C_E} + \frac{1}{q_{E,max}}$

- The Freundlich model which has the form $q_E = K \cdot C_E^n$ (with two fitting parameters K and n) which can be linearized into the form $log q_E = n \cdot log C_E + log K$

In the table below, I've given you some C_E and q_E data (along with the transformed data for the two models) for the adsorption of an organic contaminant from water onto granular activated carbon.

C_E	q_E	$1/C_E$	$1/q_E$	$\log(C_E)$	$\log(q_E)$
0.01	4.1	100	0.244	-2.000	0.613
0.02	4.5	50	0.222	-1.699	0.653
0.04	5.2	25	0.192	-1.398	0.716
0.08	6.2	12.5	0.161	-1.097	0.792
0.16	7	6.25	0.143	-0.796	0.845
0.32	7.8	3.125	0.128	-0.495	0.892
0.64	9.2	1.563	0.109	-0.194	0.964
1.28	10.6	0.781	0.094	0.107	1.025

Using the ANOVA tables on the following page, state which one of the two adsorption models is most appropriate for the given data and report the numerical values of the two best-fit constants for that model.

ANOVA table from regression analysis for linearized form of Langmuir model:

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.895731738
R Square	0.802335347
Adjusted R Square	0.769391238
Standard Error	0.025784599
Observations	8

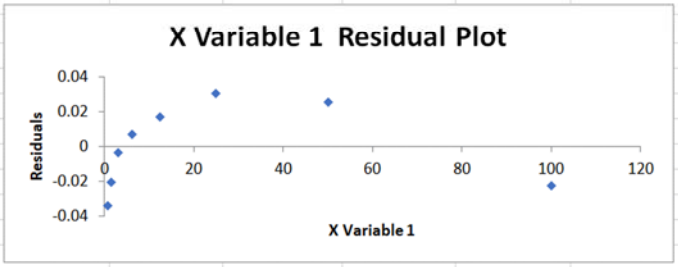
ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	1	0.016191941	0.016191941	24.35444079	0.002616979
Residual	6	0.003989073	0.000664846		
Total	7	0.020181014			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	0.127095968	0.011504396	11.04760041	3.27305E-05	0.098945725	0.15524621	0.098945725	0.15524621
X Variable 1	0.001390695	0.000281801	4.935021863	0.002616979	0.000701152	0.002080237	0.000701152	0.002080237

RESIDUAL OUTPUT

Observation	*Predicted Y*	*Residuals*
1	0.266165449	-0.022263009
2	0.196630708	0.025591514
3	0.161863338	0.030444355
4	0.144479653	0.01681067
5	0.13578781	0.007069333
6	0.131441889	-0.003236761
7	0.129268928	-0.020573276
8	0.128182448	-0.033842825



ANOVA table from regression analysis for linearized form of Freundlich model:

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.998489271
R Square	0.996980825
Adjusted R Square	0.996477629
Standard Error	0.00870372
Observations	8

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	1	0.15009268	0.15009268	1981.297681	8.61008E-09
Residual	6	0.000454528	7.57547E-05		
Total	7	0.150547209			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	1.000524094	0.00522462	191.5017978	1.36798E-12	0.98773991	1.013308279	0.98773991	1.013308279
X Variable 1	0.198584496	0.004461393	44.51177014	8.61008E-09	0.187667861	0.209501131	0.187667861	0.209501131

RESIDUAL OUTPUT

Observation	*Predicted Y*	*Residuals*
1	0.603355103	0.009428754
2	0.663134993	-0.009922479
3	0.722914882	-0.006911539
4	0.782694772	0.009696917
5	0.842474662	0.002623378
6	0.902254552	-0.01015995
7	0.962034442	0.001753385
8	1.021814332	0.003491533

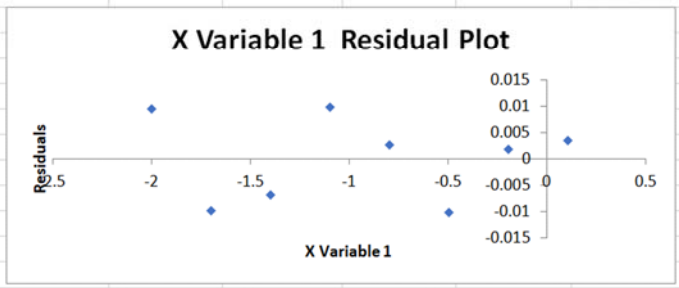


Table A.1

Specific Gravities of Selected Engineering Materials

(a) Common Manometer Liquids at 20°C

Liquid	Specific Gravity
E.V. Hill blue oil	0.797
Meriam red oil	0.827
Benzene	0.879
Dibutyl phthalate	1.04
Monochloronaphthalene	1.20
Carbon tetrachloride	1.595
Bromoethylbenzene (Meriam blue)	1.75
Tetrabromoethane	2.95
Mercury	13.55

Source: Data from References [1–3].

Material	Specific Gravity (—)
Aluminum	2.64
Balsa wood	0.14
Brass	8.55
Cast Iron	7.08
Concrete (cured)	2.4[a]
Concrete (liquid)	2.5[a]
Copper	8.91
Ice (0°C)	0.917
Lead	11.4
Oak	0.77
Steel	7.83
Styrofoam (1 pcf[b])	0.0160
Styrofoam (3 pcf)	0.0481
Uranium (depleted)	18.7
White pine	0.43

Source: Data from Reference [4].
[a]depending on aggregate.
[b]pounds per cubic foot.

Table A.6

Thermodynamic Properties of Common Gases at STP[a]

Gas	Chemical Symbol	Molecular Mass, M_m	R^b $\left(\frac{J}{kg \cdot K}\right)$	c_p $\left(\frac{J}{kg \cdot K}\right)$	c_v $\left(\frac{J}{kg \cdot K}\right)$	$k = \frac{c_p}{c_v}$ (–)
Air	—	28.98	286.9	1004	717.4	1.40
Carbon dioxide	CO_2	44.01	188.9	840.4	651.4	1.29
Carbon monoxide	CO	28.01	296.8	1039	742.1	1.40
Helium	He	4.003	2077	5225	3147	1.66
Hydrogen	H_2	2.016	4124	14,180	10,060	1.41
Methane	CH_4	16.04	518.3	2190	1672	1.31
Nitrogen	N_2	28.01	296.8	1039	742.0	1.40
Oxygen	O_2	32.00	259.8	909.4	649.6	1.40
Steam[c]	H_2O	18.02	461.4	~2000	~1540	~1.30

Source: Data from References [7, 16, 17].
[a]STP = standard temperature and pressure, $T = 15°C = 59°F$ and $p = 101.325$ kPa (abs) = 14.696 psia.
[b]$R \equiv R_u/M_m$; $R_u = 8314.3$ J/(kgmol · K) = 1545.3 ft · lbf/(lbmol · °R); 1 Btu = 778.2 ft · lbf.
[c]Water vapor behaves as an ideal gas when superheated by 55°C (100°F) or more.

Table A.2

Physical Properties of Common Liquids at 20°C

Liquid	Isentropic Bulk Modulus[a] (GN/m²)	Specific Gravity (—)
Benzene	1.48	0.879
Carbon tetrachloride	1.36	1.595
Castor oil	2.11	0.969
Crude oil	—	0.82–0.92
Ethanol	—	0.789
Gasoline	—	0.72
Glycerin	4.59	1.26
Heptane	0.886	0.684
Kerosene	1.43	0.82
Lubricating oil	1.44	0.88
Methanol	—	0.796
Mercury	28.5	13.55
Octane	0.963	0.702
Seawater[b]	2.42	1.025
SAE 10W oil	—	0.92
Water	2.24	0.998

Source: Data from References [1, 5, 6].
[a]Calculated from speed of sound; 1 GN/m² = 10^9 N/m² (1 N/m² = 1.45×10^{-4} lbf/in.²).
[b]Dynamic viscosity of seawater at 20°C is $\mu = 1.08 \times 10^{-3}$ N · s/m². (Thus, the kinematic viscosity of seawater is about 5 percent higher than that of freshwater.)

Table A.8

Properties of Water (SI Units)

Temperature, T (°C)	Density, ρ (kg/m³)	Dynamic Viscosity, μ (N · s/m²)	Kinematic Viscosity, ν (m²/s)	Surface Tension, σ (N/m)	Vapor Pressure, p_v (kPa)	Bulk Modulus, E_v (GPa)
0	1000	1.76E-03	1.76E-06	0.0757	0.661	2.01
5	1000	1.51E-03	1.51E-06	0.0749	0.872	
10	1000	1.30E-03	1.30E-06	0.0742	1.23	
15	999	1.14E-03	1.14E-06	0.0735	1.71	
20	998	1.01E-03	1.01E-06	0.0727	2.34	2.21
25	997	8.93E-04	8.96E-07	0.0720	3.17	
30	996	8.00E-04	8.03E-07	0.0712	4.25	
35	994	7.21E-04	7.25E-07	0.0704	5.63	
40	992	6.53E-04	6.59E-07	0.0696	7.38	
45	990	5.95E-04	6.02E-07	0.0688	9.59	
50	988	5.46E-04	5.52E-07	0.0679	12.4	2.29
55	986	5.02E-04	5.09E-07	0.0671	15.8	
60	983	4.64E-04	4.72E-07	0.0662	19.9	
65	980	4.31E-04	4.40E-07	0.0654	25.0	
70	978	4.01E-04	4.10E-07	0.0645	31.2	
75	975	3.75E-04	3.85E-07	0.0636	38.6	
80	972	3.52E-04	3.62E-07	0.0627	47.4	
85	969	3.31E-04	3.41E-07	0.0618	57.8	
90	965	3.12E-04	3.23E-07	0.0608	70.1	2.12
95	962	2.95E-04	3.06E-07	0.0599	84.6	
100	958	2.79E-04	2.92E-07	0.0589	101	

THE END!